OPTIMAL LOGISTIC CONTAINER PACKING USING REINFORCEMENT LEARNING CONTROLLED 5 DOF PICK AND PLACE ARM

A PROJECT REPORT

Submitted by

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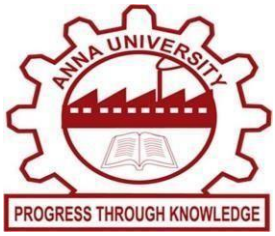
in partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

in

MECHANICAL ENGINEERING



DEPARTMENT OF PRODUCTION TECHNOLOGY

MADRAS INSTITUTE OF TECHNOLOGY

ANNA UNIVERSITY: CHENNAI 600 044

MAY 2025

A note on this web edition (2026)

This is the B.E. project report submitted to Anna University in May 2025, edited in 2026 for publication on my website. Everything not listed below is unchanged from the submitted version.

1. Student register numbers have been removed.

2. Table 8.1 originally presented the published benchmark results of GOPT, the method our policy was built on, with GOPT labelled "Ours". These are the GOPT authors' figures, not our measurements. The table, the text around it and the conclusion have been relabelled.

3. The physical figures in Table 8.2 and Section 8.2.4 (68.7% packing efficiency, 12 items per minute, a positioning error of 2 mm, 73.3% and 67.5%) are not backed by surviving logs, and 12 items per minute is inconsistent with the robot's planned cycle of about 16 s per box. They are marked as unverified. The trial case in Section 8.3 is kept as submitted; its logs were lost with the project laptop, so it could not be re-checked.

4. In 2026 I analysed the one policy checkpoint that survived and found that it had not learned: a weight initialisation bug left its choices effectively random among feasible placements. The retrained, robot-aware policy and its results are described in the OverPack technical report (2026), published alongside this report.

5. Some explanatory figures in the background chapters are reproduced from the published papers that describe those methods, and are not our own work.

Anush Bharathi R

BONAFIDE CERTIFICATE

Certified that this project report entitled **"OVERPACK: OPTIMAL LOGISTIC CONTAINER PACKING USING REINFORCEMENT LEARNING CONTROLLED 5 DOF PICK AND PLACE ARM"** is the bonafide work of **ANUSH BHARATHI R, KAMRAN B, NILAVAN KJ, and SHIBI PRAVEEN S**, who carried out the work under my supervision.

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ACKNOWLEDGMENT

It is a pleasure to place on record our profound sense of gratitude to our project guide **Dr. P. KARTHIKEYAN**, Assistant Professor, Department of Production Technology, Madras Institute of Technology, Chennai for his valuable guidance, advice, motivation and encouragement which helped us immensely to accomplish this project successfully.

We wish to express our sincere thanks to our project coordinator **Dr. P. GANESH**, Associate Professor and the panel members **Dr. N. SRIRANGARAJALU**, Assistant Professor, **Dr. A. VISAGAN**, Teaching Fellow and **Mr. P. MANI**, Teaching Fellow and our Faculty Advisor, Department of Production Technology, Madras Institute of Technology, Chennai for their valuable suggestions and constant encouragement.

We thank **Dr. J. JANCIRANI**, Professor and Head of the Department of Production Technology for extending facilities for carrying out the project work. We are highly thankful to **Dr. K. RAVICHANDRAN**, Dean, Madras Institute of Technology, Chennai for providing the necessary facilities to carry out the project.

We are grateful to all the staff members and the Department of Production Technology, MIT Campus, Anna University, Chennai, India for helping to complete the project.

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ABSTRACT

The logistics industry plays a critical role in global commerce, with container packing being a vital aspect of optimizing operational efficiency. The project titled "OPTIMAL LOGISTIC CONTAINER PACKING USING REINFORCEMENT LEARNING CONTROLLED 5 DOF PICK AND PLACE ARM" addresses the challenges in container packing, an NP-hard combinatorial problem that impacts fuel efficiency, operational costs, and environmental sustainability.

This study leverages advancements in artificial intelligence, specifically Reinforcement Learning (RL), to design an automated packing system utilizing a 5-DOF robotic arm. The RL framework integrates a dynamic learning model with heuristic rules, optimizing item placement within containers while accounting for stability, collision avoidance, and space utilization. By employing the GOPT methodology, which incorporates Transformer-based deep reinforcement learning, the system achieves enhanced generalization and adaptability across containers of varying dimensions.

The project successfully demonstrates the integration of RL algorithms with robotic systems, providing an innovative solution to logistics inefficiencies. This work contributes to advancing sustainable logistics practices and paves the way for future automation in container packing.

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CHAPTER 1
INTRODUCTION

The modern logistics and supply chain industry is a critical backbone for global commerce, involving the movement of goods across various distances and through multiple stages. Within this intricate network, container packing holds paramount importance, as it directly impacts operational efficiency, costs, and service quality. Optimal packing of goods into containers ensures maximum space utilization, reduces transportation costs, and meets delivery deadlines. However, achieving optimal packing is a complex challenge due to various constraints, such as item shapes, sizes, fragility, and the sequence of loading and unloading.

1.1 THE CONTAINER PACKING PROBLEM

The container packing problem (CPP) is a well-known combinatorial optimization challenge that involves arranging objects within a container to maximize space utilization while adhering to practical constraints. Traditional approaches to solving CPP have focused on heuristic and metaheuristic algorithms that aim to find near-optimal solutions. These methods, although effective in specific scenarios, often fall short when applied to real-world situations that involve dynamic and unpredictable variables.

The complexity of CPP is further exacerbated by the need to consider factors such as stability, ease of loading and unloading, and the sequence in which items must be accessed. These factors are crucial in ensuring the safety of goods and the efficiency of the packing process. Traditional methods often overlook these practical considerations, leading to suboptimal solutions and increased operational costs.

1.2 IMPACT OF EFFICIENT CONTAINER PACKING ON LOGISTICS

Efficient container packing directly influences several logistical factors, including fuel efficiency, cargo safety, and overall operational costs. By optimizing the arrangement of items within containers, it is possible to reduce the number of trips required for transportation, thus conserving fuel and reducing emissions. Additionally, well-planned packing strategies minimize the risk of damage to goods, ensuring their safe delivery.

Research in this area has shown that adopting advanced packing techniques can significantly enhance space utilization and streamline the loading and unloading processes. For example, integrating intelligent algorithms with automated systems, such as robotic arms, can lead to more precise and efficient packing solutions, ultimately benefiting the entire supply chain.

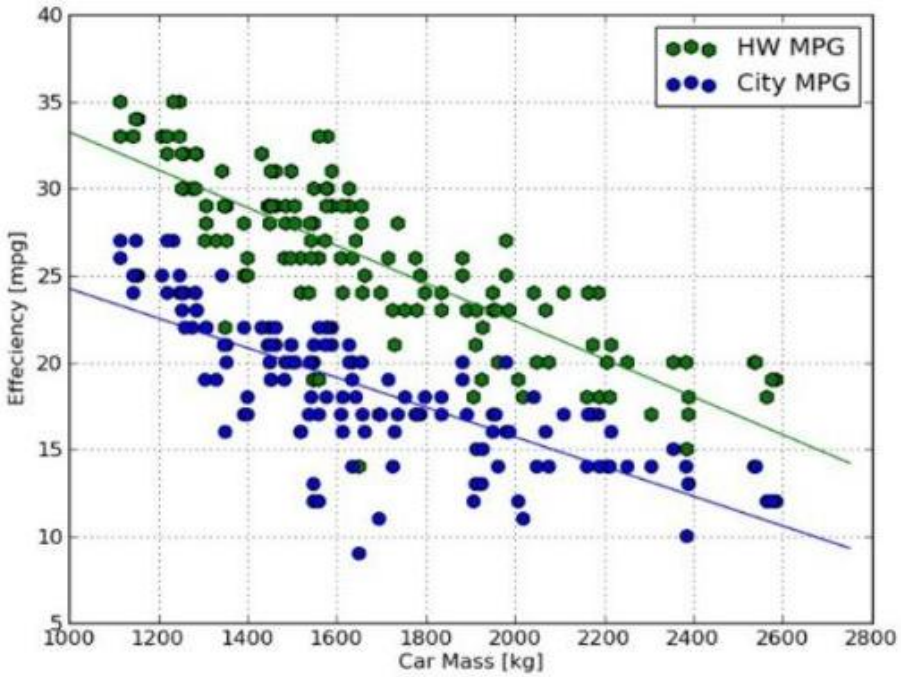


Figure 1.1 Efficiency of container packing on logistics

1.3 INTEGRATION OF REINFORCEMENT LEARNING AND ROBOTIC SYSTEMS

Recent advancements in artificial intelligence (AI) and machine learning (ML) have opened new avenues for addressing complex optimization problems. Reinforcement learning (RL), a subset of ML, has shown great promise in solving dynamic and complex problems by learning optimal policies through trial and error. RL algorithms can adapt to changing environments, making them well-suited for applications in logistics and container packing.

The integration of RL with robotic systems, such as a 5 Degree of Freedom (DOF) Pick and Place arm, introduces a novel approach to solving the container packing problem. A 5 DOF arm provides flexibility and precision in manipulating objects, making it ideal for arranging items within a confined space. By leveraging RL, the pick and place arm can learn to optimize the packing process, taking into account factors such as item shapes, sizes, and stability.

The 5 DOF Pick and Place arm is capable of executing complex movements, including translation and rotation along multiple axes. This capability allows for the precise placement of items to maximize container space utilization. When controlled by an RL algorithm, the arm can dynamically adapt to different packing scenarios, continuously improving its strategy through interaction with the environment.

About 65.4% of all goods in India are transported through the roadways, therefore, making it very crucial for the economy. The transportation sector in India emits an estimated 261 tonnes of CO2, of which 94.5% is contributed by road transports. Efficient packing plans for the trucking industry can help to reduce its carbon footprint, protect goods from damage, and also save on shipping costs.

CHAPTER 2
LITERATURE REVIEW

G. Wäscher et al., (2007), The paper titled "An improved typology of cutting and packing problems" established a comprehensive classification system for cutting and packing problems. This work provided clear problem definitions and categorizations that have guided subsequent research in the field, creating a standardized framework for addressing various types of packing challenges.

S. Martello et al., (2000), The paper titled "The three-dimensional bin packing problem" formalized many of the mathematical models still used today for 3D bin packing. The authors provided rigorous problem definitions and developed exact and approximate algorithms, establishing foundational approaches that have influenced decades of subsequent research.

T. G. Crainic et al., (2008), The paper titled "Extreme point-based heuristics for three-dimensional bin packing" identified strategic positions within bins where items could be placed, significantly improving computational efficiency. This work addressed the challenge of identifying feasible packing positions quickly in complex 3D environments through an innovative extreme-point representation.

K. Karabulut and M. M. İnceoglu (2005). The paper titled "A hybrid genetic algorithm for packing in 3D with deepest bottom left with fill method" introduced a hybrid genetic algorithm that integrates heuristic strategies for 3D bin packing. The proposed approach improved efficiency by combining evolutionary techniques with structured heuristic methods, achieving better space utilization and stability in packed arrangements.

C. T. Ha et al., (2017), The paper titled "An online packing heuristic for the three-dimensional container loading problem in dynamic environments and the Physical Internet" presented an adaptive heuristic for online container loading that accounts for real-time changes in dynamic logistics scenarios. The approach enhances flexibility and efficiency in automated packing under variable constraints.

L. Wang et al., (2010), The paper titled "Two natural heuristics for 3D packing with practical loading constraints" introduced heuristic-based strategies tailored to real-world packing conditions. These methods focus on maximizing space utilization while ensuring the practicality of loading and handling constraints in automated packing systems.

H. Hu et al., (2017), The paper titled "Solving a new 3D bin packing problem with deep reinforcement learning method" applied deep reinforcement learning (DRL) techniques to 3D bin packing, demonstrating the potential of AI-driven approaches in optimizing space allocation. The study highlights the advantages of learning-based strategies in tackling complex packing environments.

L. Duan et al., (2018), The paper titled "A multi-task selected learning approach for solving 3D flexible bin packing problem" introduces a reinforcement learning framework that employs multi-task selected learning to optimize flexible bin packing solutions. The approach enhances adaptability in dynamic packing environments by leveraging multiple learning tasks to improve policy generalization.

A. Laterre et al., (2018), The paper titled "Ranked reward: Enabling self-play reinforcement learning for combinatorial optimization" presents a ranked

reward mechanism that enables self-play reinforcement learning for solving complex combinatorial optimization problems, such as bin packing. This method improves learning efficiency and solution quality by dynamically adjusting reward structures based on performance rankings.

H. Zhao et al., (2021), The paper titled "Online 3D bin packing with constrained deep reinforcement learning" formulates the online bin packing problem as a constrained deep reinforcement learning (DRL) task. The study incorporates feasibility constraints to ensure practical applicability, yielding improved packing stability and execution feasibility in dynamic warehouse settings.

H. Zhao et al., (2022), The paper titled "Learning practically feasible policies for online 3D bin packing" extends prior work by integrating constraint-based policy learning within DRL models, enhancing real-world implementation feasibility. The framework ensures practical stability and efficient space utilization across diverse packing environments.

R. Verma et al., (2020), The paper titled "A generalized reinforcement learning algorithm for online 3D bin-packing" introduces a flexible reinforcement learning framework capable of handling various packing configurations. The approach generalizes across different item and bin sizes, improving adaptability in warehouse automation.

Z. Yang et al., (2021), The paper titled "Packerbot: Variable-sized product packing with heuristic deep reinforcement learning" combines heuristic methods with DRL to develop an adaptive bin packing system. The study highlights the benefits of heuristic-driven rewards, enhancing DRL agent performance and overall packing efficiency.

Y. Jiang et al., (2021), The paper titled "Learning to solve 3-D bin packing problem via deep reinforcement learning and constraint programming" integrates constraint programming into a DRL framework, improving space efficiency and packing accuracy. The hybrid approach demonstrates robustness in handling complex packing scenarios.

D. Li et al., (2022). The paper titled "One model packs thousands of items with recurrent conditional query learning" introduces a novel learning framework that utilizes recurrent conditional query learning to optimize large-scale 3D bin packing. The approach enables efficient item placement across diverse bin configurations and enhances computational efficiency.

H. Xiong et al., (2024), The paper titled "GOPT: Generalizable online 3D bin packing via transformer-based deep reinforcement learning" presents a reinforcement learning framework for online bin packing, incorporating a Placement Generator and a Packing Transformer. The study highlights the model's ability to generalize across varying bin dimensions, demonstrating superior performance compared to existing methods.

H. Xiong et al., (2023), The paper titled "Towards reliable robot packing system based on deep reinforcement learning" explores the application of DRL-based bin packing in real-world robotic systems, focusing on stability, feasibility constraints, and deployment challenges in automated warehouses.

T. Kim and J. Kim (2023), The paper titled "Voxel Transformer with Density-Aware Deformable Attention for 3D Object Detection" introduces a Voxel Transformer designed for 3D object detection. The approach leverages density-aware deformable attention mechanisms, enhancing object recognition accuracy in cluttered environments.

R. Shome et al., (2019), The paper titled "Towards robust product packing with a minimalistic end-effector" investigates robotic packing strategies, emphasizing the role of minimalistic end-effectors in achieving reliable product handling. The study offers insights into reducing complexity while maintaining placement precision.

F. Wang and K. Hauser (2019). The paper titled "Stable bin packing of non-convex 3D objects with a robot manipulator" proposes techniques for packing non-convex objects, addressing stability challenges in robotic manipulation through optimization strategies.

M. Agarwal et al., (2020), The paper titled "Jampacker: An efficient and reliable robotic bin packing system for cuboid objects" presents a reinforcement learning-based robotic bin packing system, demonstrating high efficiency in handling cuboid-shaped items with improved stability and adaptability.

N. Park (2023), The paper titled "BoxStacker: Deep reinforcement learning for 3D bin packing problem in virtual environment of logistics systems" presents a DRL-based packing framework tailored for logistics simulations. The study leverages deep reinforcement learning to optimize item placement in dynamic environments, demonstrating improved efficiency in virtual logistics operations.

X. Que et al., (2023), The paper titled "I Have a DREAM to Pack 3D Boxes: Divide-and-Conquer for 3D Bin Packing with Transformers" introduces a novel divide-and-conquer strategy for 3D bin packing using Transformer-based models. The method partitions the bin into smaller sub-regions, efficiently handling complex packing scenarios through hierarchical decision-making.

H. Zhao et al., (2022), The paper titled "Learning efficient online 3D bin packing on packing configuration trees" develops an approach based on Packing

Configuration Trees (PCT) to enhance online 3D bin packing. The study integrates heuristic search rules with deep reinforcement learning to improve space utilization while maintaining computational efficiency.

2.1 NEED FOR STUDY

The logistics industry, a critical driver of economic growth, is currently facing significant challenges. Studies suggest that optimizing truck utilization could potentially reduce CO_2 emissions by 185 million tonnes annually by 2050, aligning with global sustainability goals. This optimization could also result in substantial economic benefits, with projected cost savings of approximately INR 85 lakh crore. These figures highlight the urgency of addressing inefficiencies within logistics operations.

Despite its vast economic value, the Indian logistics sector remains largely unorganized, with only 10-15% employing modern systems. Container utilization rates are inefficient, averaging between 60-75%, and the time required for loading a single container can extend up to 3-4 hours.

The integration of robotics and artificial intelligence offers a promising solution. This study focuses on leveraging a 5-DOF robotic arm powered by Reinforcement Learning (RL) to develop an advanced, automated packing system.

2.2 OVERVIEW OF THE PROBLEM

The logistics industry faces a critical challenge in optimizing container packing, an NP-hard combinatorial problem. Inefficient packing impacts operational costs, CO_2 emissions, and resource utilization, with global implications for e-commerce and sustainability. Despite technological advances, the integration of automation remains limited, particularly in the Indian logistics sector, where manual packing methods dominate, leading to low container utilization rates, extended packing times, and environmental inefficiencies.

2.3 TRADITIONAL APPROACHES TO BIN PACKING

Historically, container packing relied on heuristic algorithms and meta-heuristic methods. While effective for offline scenarios, these approaches lack adaptability to dynamic, real-time online packing problems where decisions must be made sequentially without knowledge of incoming items. As highlighted in studies such as Ha et al., [4] and Wang et al., [16], heuristic methods are computationally efficient but struggle to generalize across varying bin configurations and item shapes.

Extreme point-based heuristics and genetic algorithms introduced improvements but remained computationally expensive for complex logistics tasks. Research by Martello et al., [11] and Crainic et al., [2] outlined limitations in scalability and adaptability, emphasizing the need for novel solutions.

2.4 REINFORCEMENT LEARNING AND TRANSFORMER-BASED INNOVATIONS

The emergence of deep reinforcement learning (DRL) marked a turning point in addressing online 3D bin packing. Algorithms like PackMan [18] and PackerBot [19] demonstrated the benefits of combining traditional heuristics with modern learning approaches. However, scalability and generalization remained key challenges.

Recent advances such as GOPT[20] introduced transformer-based architectures, achieving high generalization capabilities across diverse bin dimensions without retraining. By leveraging self-attention mechanisms, GOPT establishes spatial correlations between items and sub-spaces, providing flexible and accurate modeling of packing constraints. Experimental validation of GOPT highlighted improvements in space utilization and packing efficiency, with adaptability across varied container types.

2.5 RESEARCH GAPS

Despite progress, several critical research gaps persist:

1. **Generalization Across Irregular Shapes**: Current methods primarily focus on cuboidal items, neglecting irregular shapes or deformable objects.
2. **Integration with Robotic Systems**: Algorithms often fail to address constraints related to motion planning, collision avoidance, and stability in physical implementations.
3. **Computational Efficiency**: High-performance algorithms require optimization for real-time adaptability in dynamic logistics scenarios.
4. **Scalability Across Dimensions**: Limited exploration of models that
5. dynamically adapt to containers with varying dimensions without retraining.
6. **Environmental Impact**: Most studies overlook the potential for reducing carbon footprints through optimized packing strategies.

These gaps underline the need for a comprehensive framework that combines offline planning with adaptive online execution. The proposed approach integrates **voxel-based transformer architectures**, multi-modal fusion techniques, and robust reinforcement learning models to address these challenges.

CHAPTER 4
CONCEPTUAL DESIGN AND CALCULATIONS

The conceptual design phase is a critical step in translating theoretical frameworks and objectives into a tangible, functional system. The design focuses on developing a 5-DOF pick-and-place robotic system controlled by Reinforcement Learning (RL). The primary aim is to ensure that the design adheres to the project's goals of efficiency, sustainability, and adaptability in a warehouse-like environment.

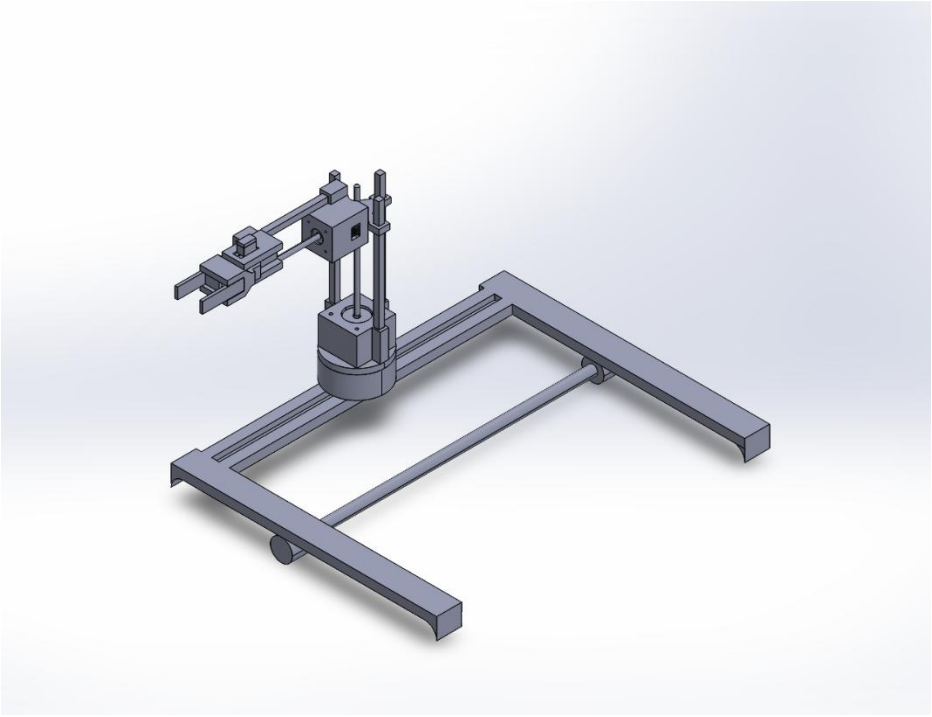


Figure 4.1 Robotic arm with gantry rail(full assembly)

4.1 ROBOTIC SYSTEM CONFIGURATION

The robotic system is designed as a 5-DOF pick-and-place arm with the following specifications:

Base Structure: A gantry-type XY rail system, providing a workspace of 1m x 0.6m. The structure is built using aluminium extrusions to ensure rigidity and modularity.

Robotic Arm: Central rotary base combined with a Z-axis motion mechanism using lead screw, offering 0.2m travel for optimal reach.

End-Effector Design: Specialized electro magnets for secure handling of items with varying dimensions.

Payload Capacity: Supports a maximum payload of 200g, scaled for experimental implementation.

Container Dimensions: Operates within a scaled container model (30cm x 20cm x 15cm) to simulate real-world logistics requirements.

4.2 COMPONENT DIMENSIONS

1. Rail

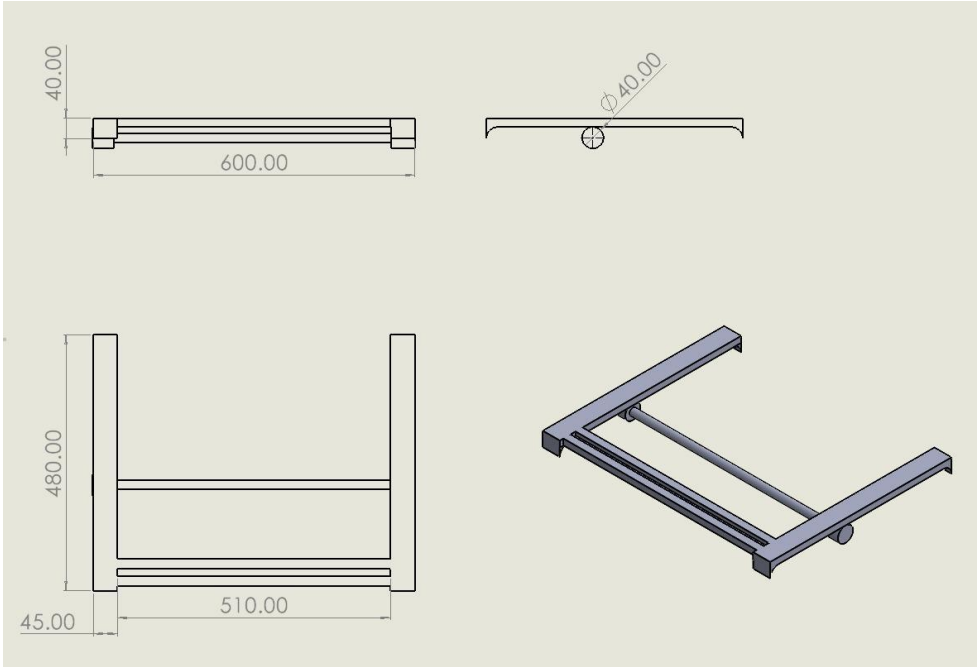


Figure 4.2 Rail

2. Base Link 1

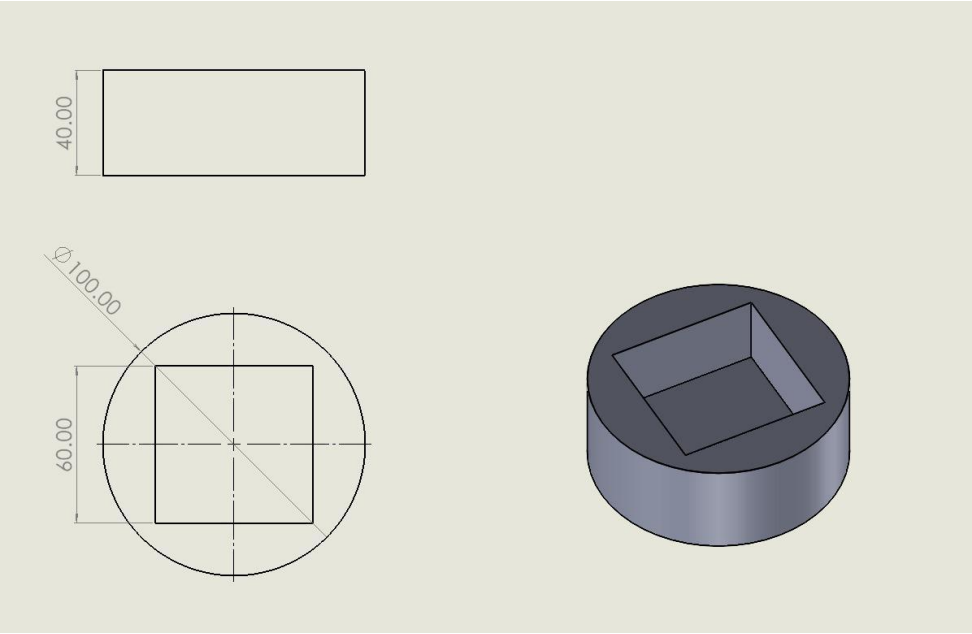


Figure 4.3 Base Link

3. Base Link 2

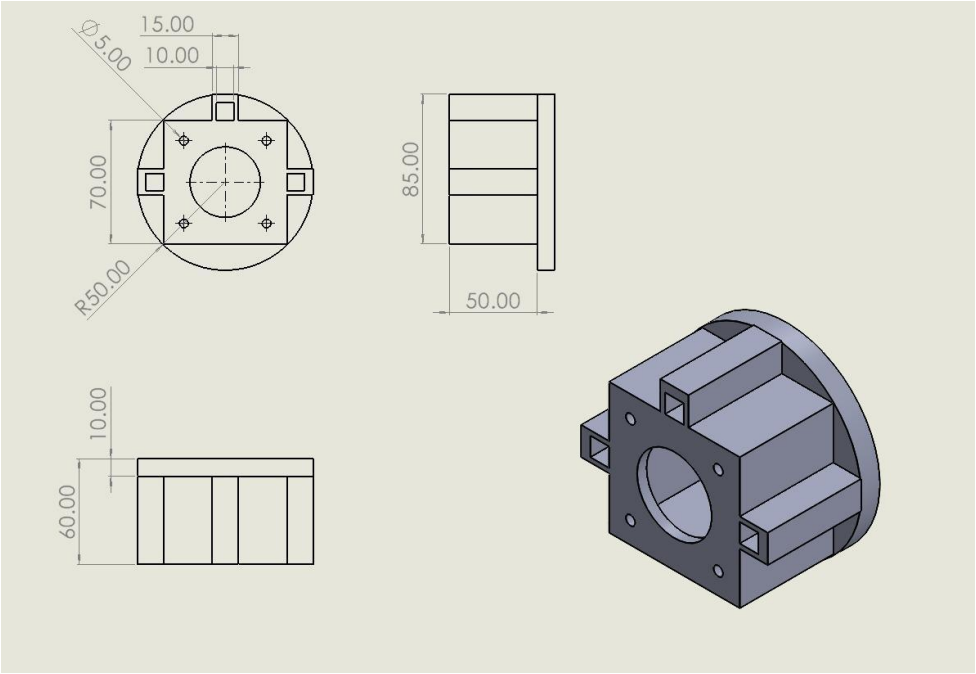


Figure 4.4 Base link 2

4. Column Link

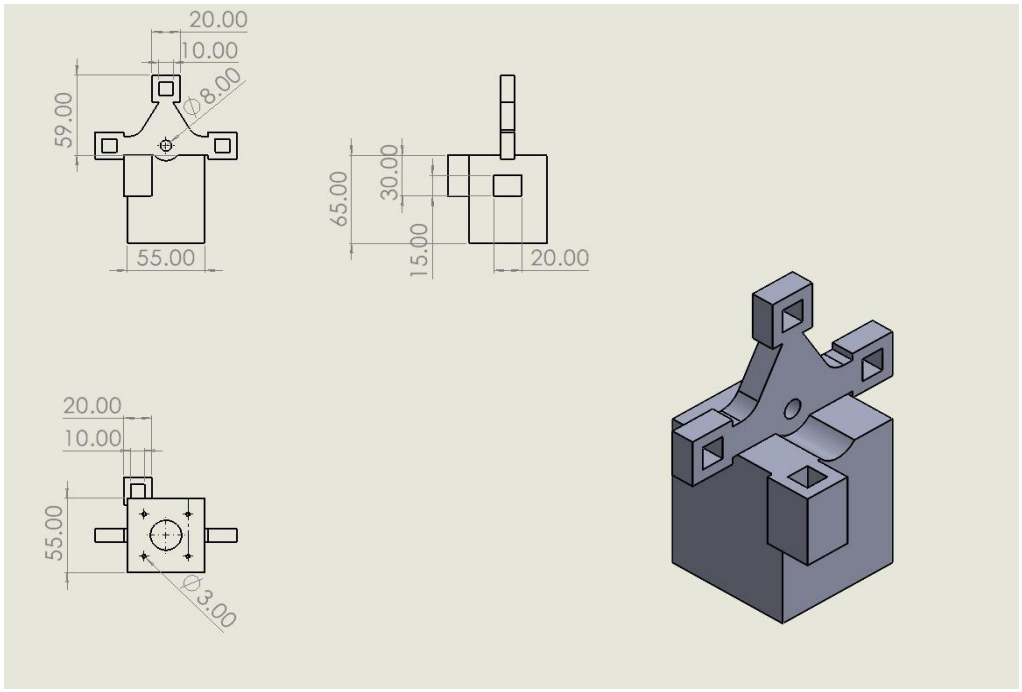


Figure 4.5 Column Link

5. Gripper Link

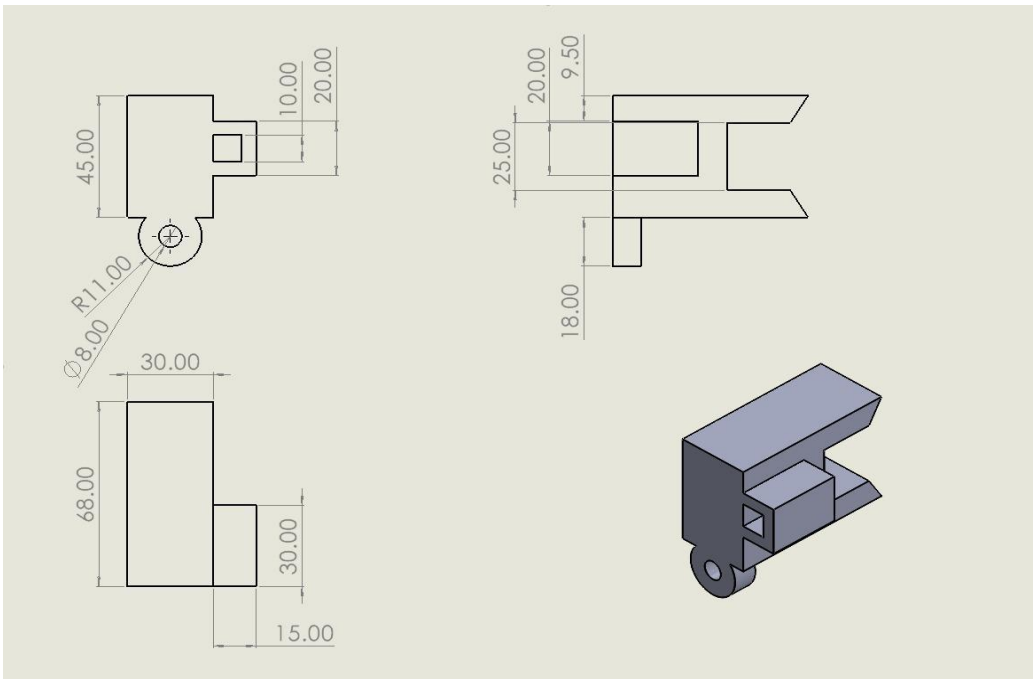


Figure 4.6 Gripper Link

6. End effector

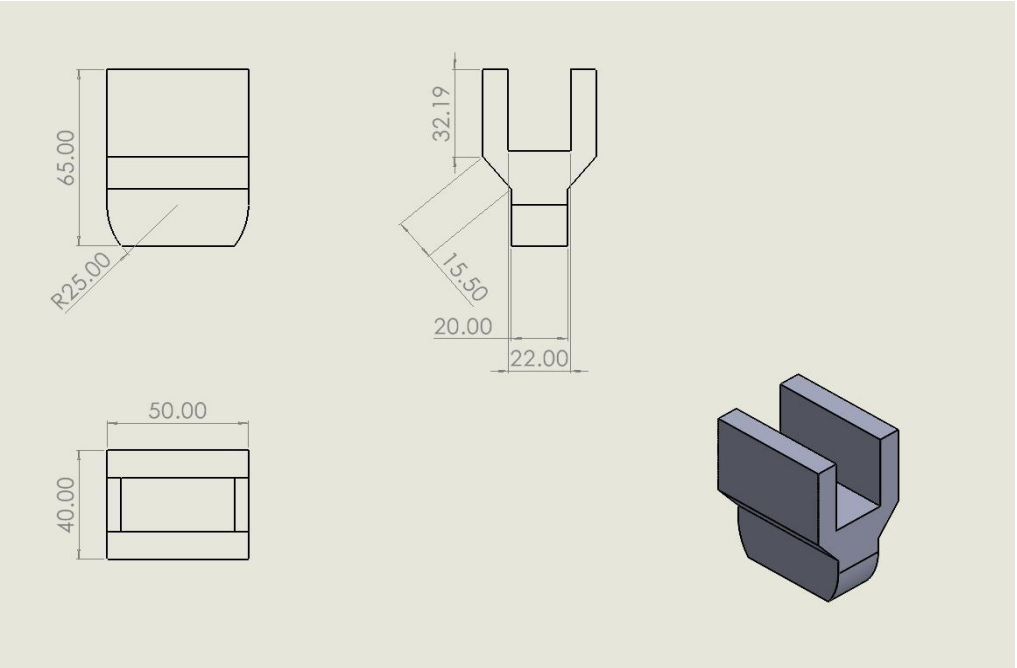


Figure 4.7 End Effector

4.3 D-H Parameters

The manipulator consists of a Cartesian base (X-Y-Z prismatic movement) followed by two additional joints: a revolute joint and a prismatic joint for vertical motion, and a final revolute joint offset by a fixed distance.

4.3.1 Robot Configuration The robotic system comprises the following joints:

- **Joint 1**: Prismatic joint translating along the X-axis
- **Joint 2**: Prismatic joint translating along the Y-axis
- **Joint 3**: Revolute joint rotating about the Z-axis
- **Joint 4**: Prismatic joint translating along the Z-axis with a horizontal offset (a4 = 0.22 m)
- **Joint 5**: Revolute joint rotating about an axis parallel to the Z-axis, located 0.25 m horizontally from Joint 4

4.3.2 DH Parameters Table

Joint (i)	θ_i (theta)	d_i (offset)	a_i (length)	α_i (twist)	Joint Type
1	0	0.45m	0	0	Prismatic
2	0	0.6m	0	π/2	Prismatic
3	360	0	0	0	Revolute
4	0	0.25m	0.22 m	0	Prismatic
5	180	0	0.25 m	0	Revolute

Table 4.1: D-H Parameters Table

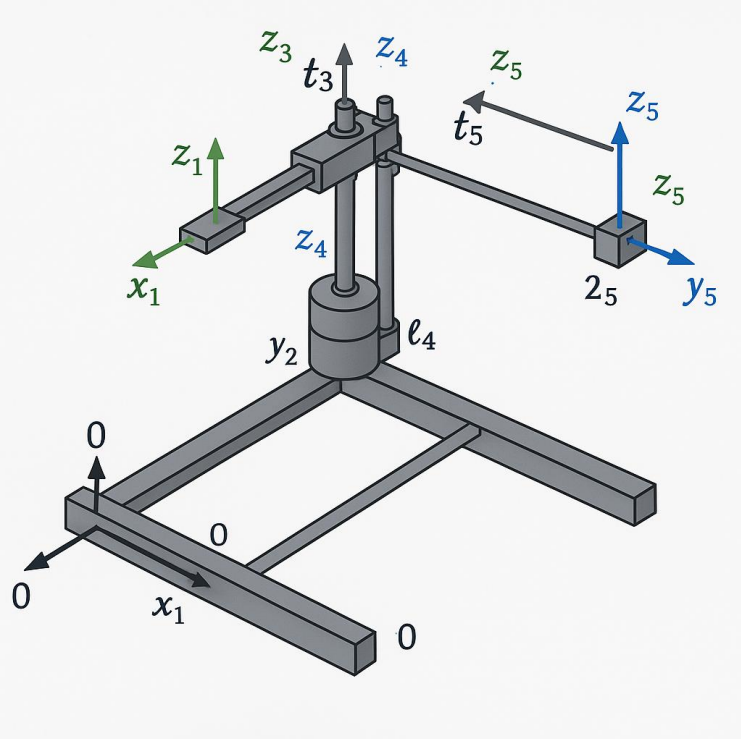


Figure 4.8 D-H Frame Assignment

4.3.3 Frame Assignments

Each coordinate frame is assigned based on the standard DH convention. Axes are placed such that:

- The Z-axis of each frame aligns with the joint axis
- The X-axis is determined by the common normal between consecutive Z-axes or is perpendicular to both

Figure 4.8 accompanying this report provides a clear 3D representation of the robotic structure with labeled frames and color-coded axes (Red: x, Green: y, Blue: z).

CHAPTER 5
ANSYS SIMULATION AND ANALYSIS

5.1 MODEL GEOMETRY AND CONFIGURATION

The CAD geometry was imported from a STEP file using the following import settings:

- Solid and surface bodies were included
- Independent parameters
- Associativity enabled
- 3D analysis type

5.2 MODEL COMPONENTS

The full assembly consisted of 33 bodies, of which 6 were activated for analysis:

- Stepper motor (NEMA23 57BYG201)
- Load-bearing lead screws
- Rail assembly
- Structural links (Link 1 and Link 2)
- MG90 servo body

This configuration provided a comprehensive representation of the main structural elements while optimizing computational resources by excluding non-essential components.

5.3 KEY DIMENSIONAL PARAMETERS

The simulation domain encompassed a volume with:

- Length X: 499.85 mm
- Length Y: 375.72 mm
- Length Z: 616.04 mm
- Total volume of active components: 4.089×10^{5} mm^3
- Total mass: 0.33734 kg

5.4 MATERIAL PROPERTIES CONFIGURATION

Three materials were assigned to different components:

5.4.1 HIGH-IMPACT ABS PLASTIC

Applied to:

- Lead screw components
- Link 1 and Link 2
- Baselink components

Properties:

- Density: 8.25×10^{-7} kg/mm^3
- Young's Modulus: 1628 MPa
- Poisson's Ratio: 0.4089
- Tensile Yield Strength: 27.44 MPa
- Tensile Ultimate Strength: 36.26 MPa
- Thermal Expansion Coefficient: 1.84×10^{-4} $°C^{-1}$

5.4.2 ALUMINUM ALLOY (6061-T6)

Applied to:

- Rail assembly

Properties:

- Density: 2.713×10^{-6} kg/mm³
- Young's Modulus: 69040 MPa
- Poisson's Ratio: 0.33
- Tensile Yield Strength: 259.2 MPa
- Tensile Ultimate Strength: 313.1 MPa
- Thermal Expansion Coefficient: 2.278×10^{-5} °C^{-1}

5.5 MESH SETTINGS

A physics-driven mesh was generated with the following parameters:

- Element size: 2.0 mm
- Adaptive sizing: Enabled
- Resolution: Default (2)
- Mesh defeaturing: Enabled
- Transition: Fast
- Element quality target: 0.05 (default)
- Smoothing: Medium

5.5.1 MESH QUALITY METRICS

The final mesh consisted of:

- 142,106 nodes
- 60,228 elements
- Element quality ranging from 0.066285 to 1.0
- Average element quality: 0.77797
- Standard deviation: 0.15843

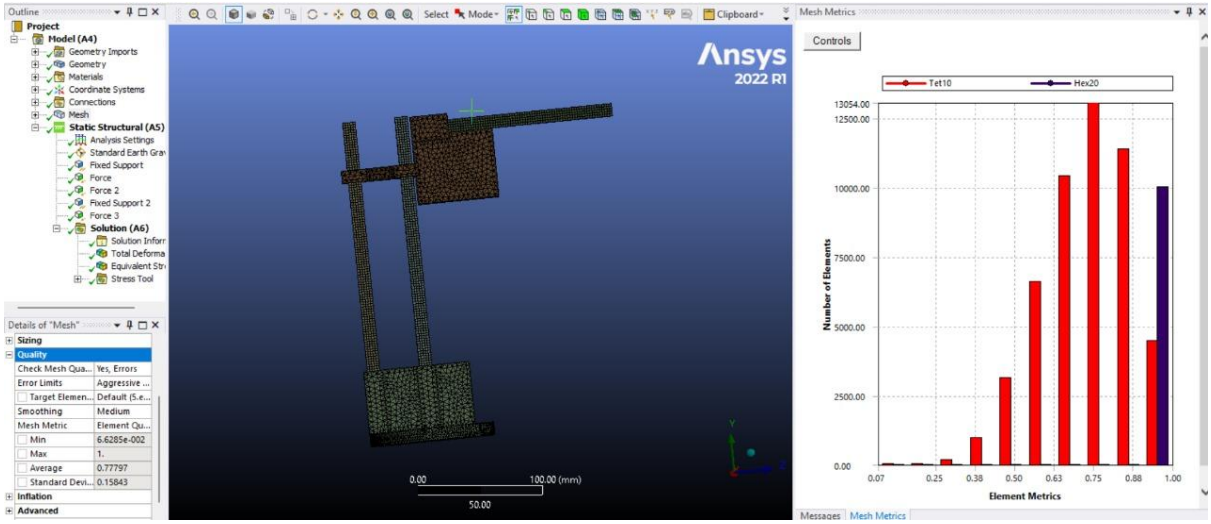


Figure 5.1 Mesh Quality Metrics

5.6 CONTACT CONFIGURATION

5.6.1 CONTACT DETECTION AND SETUP

Five active contact regions were defined using bonded contact formulation:

1. Contact Region 39: Between baselink up and lead screw (5 faces)
2. Contact Region 41: Between lead screw components (suppressed)
3. Contact Region 44: Between lead screw components (1 face)
4. Contact Region 46: Between lead screw and Link 1 (4 faces)
5. Contact Region 48: Between lead screw components (4 faces)

All contacts were configured with:

- Automatic scope mode
- Program-controlled behavior
- Default penetration tolerance
- No geometric correction applied

5.7 LOAD APPLICATION AND BOUNDARY CONDITIONS

5.7.1 GRAVITATIONAL LOAD

Standard Earth gravity (9806.6 mm/s²) was applied in the negative Y direction across all bodies to simulate real-world operating conditions.

5.7.2 APPLIED FORCES

Three strategic forces were applied to simulate operational loading:

Force 1: 6.0 N downward force (Y-component) applied to a critical surface

Force 2: 2.5 N downward force (Y-component) applied to simulate payload

Force 3: 2.5 N downward force (Y-component) applied to the extended arm

5.7.3 CONSTRAINTS

Two fixed support constraints were applied to secure the model:

1. Fixed Support 1: Applied to base connection surface
2. Fixed Support 2: Applied to secondary mounting surface

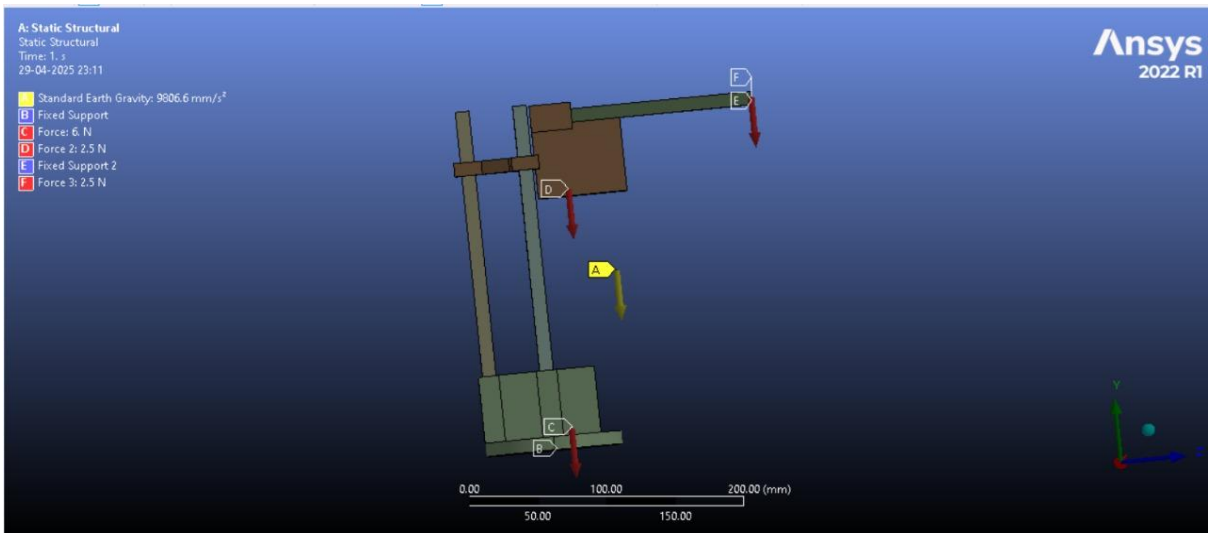


Figure 5.2 Loads and Boundary Conditions

5.8 ANALYSIS SETTINGS

5.8.1 SOLUTION CONFIGURATION

- Analysis type: Static Structural
- Single-step analysis with 1.0 second duration
- Large deflection effects: Disabled
- Environment temperature: 22°C

5.8.2 SOLVER PARAMETERS

- Solver type: Program controlled
- Newton-Raphson option: Program controlled
- Weak springs: Disabled
- Result storage: All time points

5.9 SIMULATION RESULTS AND INTERPRETATION

5.9.1 DEFORMATION ANALYSIS

The total deformation results revealed:

- Maximum deformation: 0.74436 mm (occurring on Lead screw component)
- Minimum deformation: 0.0 mm (at fixed constraints)
- Average deformation: 0.18669 mm

The deformation pattern indicated acceptable rigidity under the applied loads, with maximum displacement well within the operational tolerance of the robotic arm assembly.

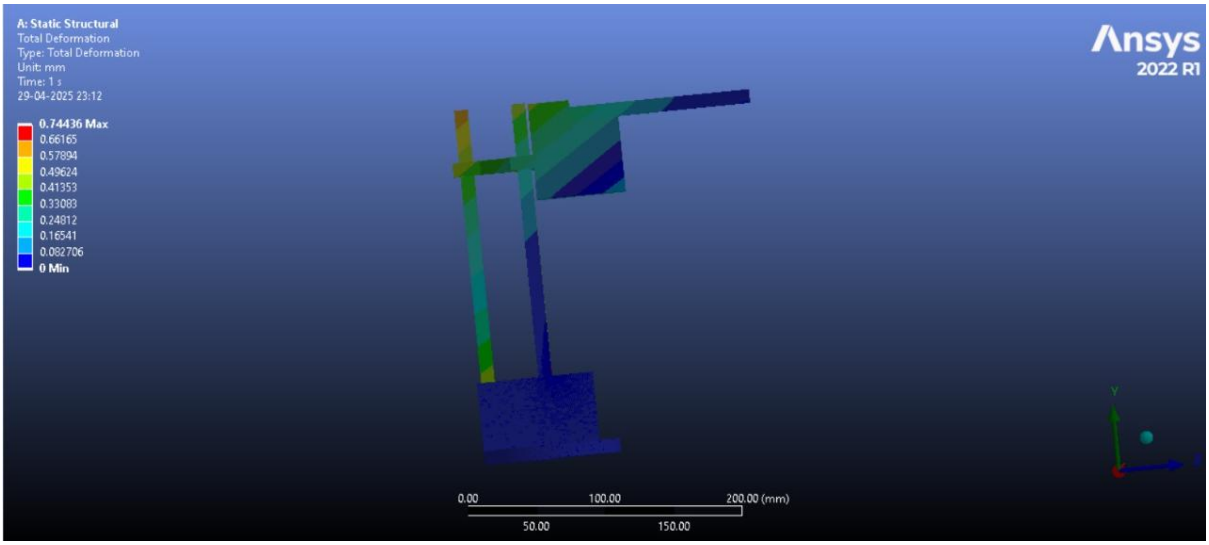


Figure 5.3 Total Deformation of Robotic Arm

5.9.2 STRESS ANALYSIS

The equivalent (von-Mises) stress results showed:

- Maximum stress: 0.87078 MPa (occurring on Lead screw 2 component)
- Minimum stress: 3.0082×10^{-12} MPa (on Lead screw component)
- Average stress: 0.03106 MPa

These stress levels are significantly below the yield strength of any materials used (27.44 MPa for ABS plastic, 250 MPa for structural steel, and 259.2 MPa for aluminum), indicating a very robust design.

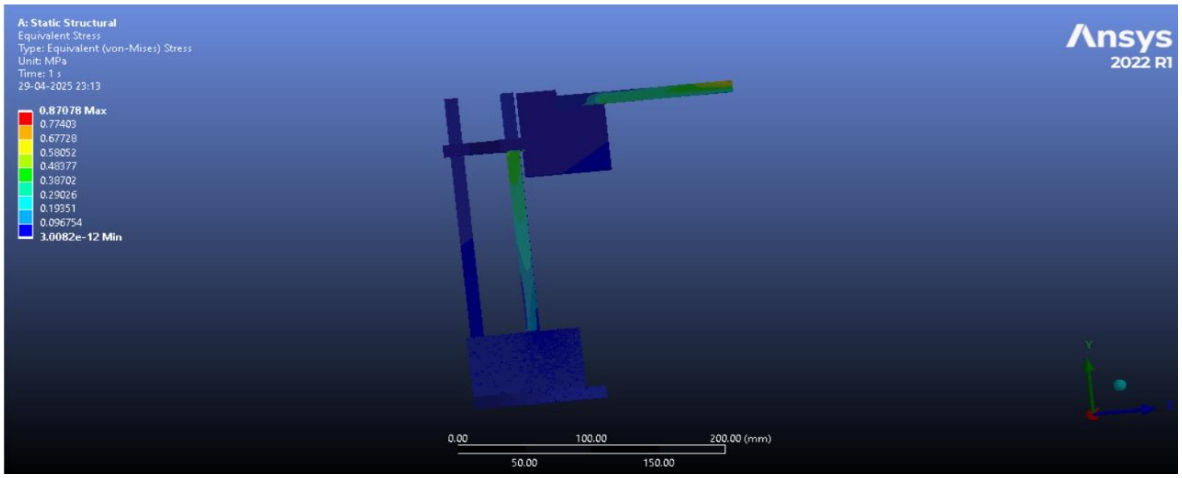


Figure 5.4 Equivalent (Von Mises) Stress acting on Robotic Arm

5.10 ANALYSIS CONCLUSIONS

The static structural analysis of the robotic arm assembly confirms that the design is structurally sound with considerable safety margins. Key observations include:

1. **Structural Integrity**: The maximum stress of 0.87078 MPa is well below the yield strengths of all materials (minimum 27.44 MPa for ABS), indicating the structure will maintain integrity under the applied loads.

2. **Deformation Characteristics**: The maximum deformation of 0.74436 mm is minimal and will not impact the precision of the robotic arm's operations.

3. **Safety Assessment**: With safety factors consistently above 15, the design demonstrates significant robustness against static loading conditions.

4. **Material Selection Validation**: The chosen materials (structural steel, ABS plastic, and aluminum alloy) are appropriate for the application, providing sufficient strength while maintaining a reasonable overall mass of 0.33734 kg.

The simulation results validate the current design parameters, suggesting that there could be opportunities for weight optimization while maintaining structural integrity.

CHAPTER 6
FABRICATION OF THE SETUP

6.1 Fabrication Process of the 5-DOF Pick and Place Robotic System

The fabrication process described here details the construction of a 5-DOF (Degrees of Freedom) pick-and-place robotic system designed for warehouse automation applications. This system was developed following rigorous conceptual design and ANSYS simulation analysis to ensure structural integrity and operational efficiency.

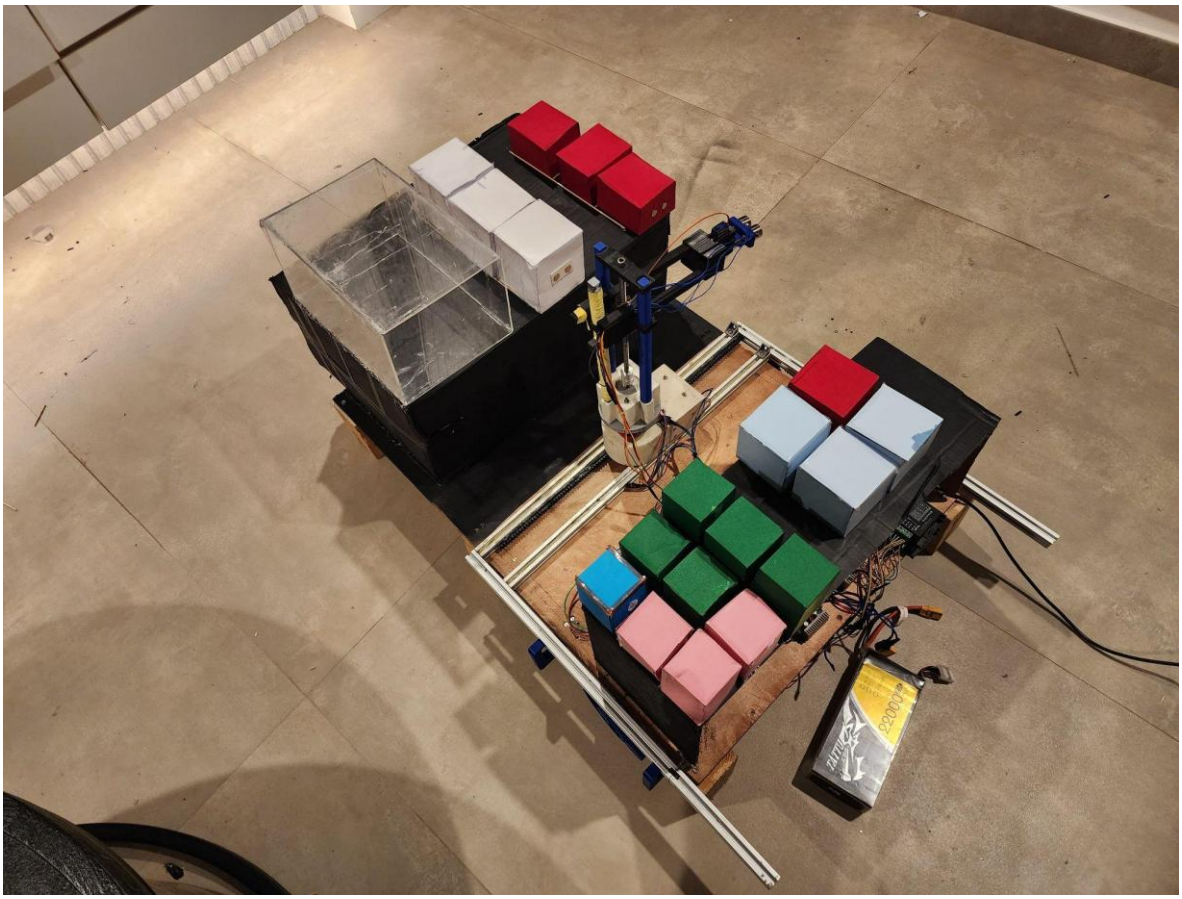


Figure 6.1 Fabricated 5-DOF Pick and Place Robot

6.2 Base Structure Construction

6.2.1 Foundation Platform

- **Materials Used:** Wooden board (approximately 600mm × 900mm) as the primary foundation
- **Process:**
    - The wooden base was cut to size and reinforced with supporting elements underneath
    - Black laminate or coating was applied to the upper surface for durability and visual contrast
    - Mounting holes were drilled at strategic locations to secure the rail system

6.2.2 Rail System Installation

- **Components Used:**
    - Aluminum extrusion rails (standard profile)
    - Linear bearings and guides
    - Timing belts and pulleys for X-Y movement
- **Assembly Process:**
    - Two parallel rails were mounted along the X-axis with precision alignment
    - Y-axis rail was installed perpendicular to the X-axis rails with mounting brackets
    - Linear bearings were installed on all rails to enable smooth motion
    - Timing belts were secured to provide precise positional control

6.3 Container/Workspace Development

6.3.1 Container Structure

- **Materials Used:**

- Black acrylic or composite material for the container walls
- Transparent acrylic for the top section to enable visibility

- **Construction Method:**
  - Container walls were cut to specifications (approximately 30cm × 20cm × 15cm)
  - Wall panels were joined using appropriate adhesives
  - Transparent acrylic top was installed to allow for visual monitoring
  - The container was mounted securely to the base platform

6.3.2 Block Organization Area

- **Components:**
  - Colored cubes (red, green, blue, white, pink) representing standardized packages
  - Black platform with defined positions for organized block placement
- **Implementation:**
  - A separate area was designated adjacent to the container for organized block storage
  - The platform was secured to the wooden base with adhesive/screws

6.4 Robotic Arm Assembly

6.4.1 Base Column Construction

- **Components Used:**
  - Blue aluminum tubing for the vertical column
  - White 3D printed or machined base mount
  - Lead screw mechanism for vertical (Z-axis) movement
  - Stepper motor for rotational movement
- **Assembly Process:**

- The base mount was secured to the Y-axis carriage
- Vertical aluminum column was installed with precise alignment
- Lead screw was installed inside the column with appropriate bearings
- Stepper motor was mounted at the base to provide rotational movement

6.4.2 Rotary Base Joint

- **Components:**
  - Circular white plastic or aluminum platform
  - Bearings for smooth rotation
  - Red mounting plate for motor connection
- **Construction Method:**
  - The rotary base was assembled with appropriate bearings
  - Mounting hardware was installed to connect to the vertical column
  - Wiring paths were created for motor and control connections

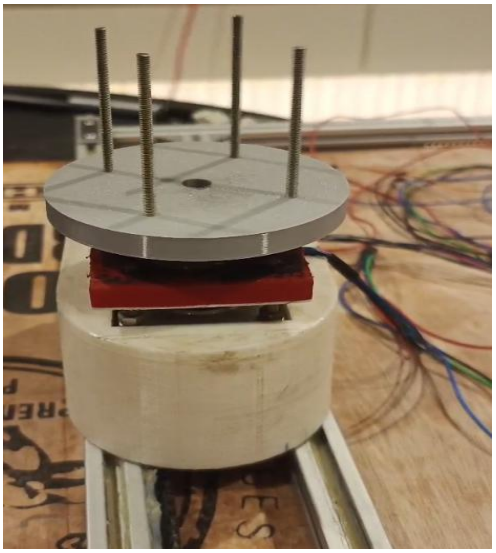


Figure 6.2 Rotary Base Joint

6.4.3 End Effector Integration

- **Components:**
  - Black horizontal arm
  - Electromagnet end effector
  - Servo motor for rotating electromagnet for holding boxes on varied dimensions
- **Assembly Process:**
  - Horizontal arm was mounted to the rotary base
  - Electromagnet was mounted at the arm tip with appropriate wiring
  - Servo motor was integrated for precise end effector control

CHAPTER 7
REINFORCEMENT LEARNING FOR 3D BIN PACKING

7.1 INTRODUCTION TO REINFORCEMENT LEARNING IN BIN PACKING

Efficient bin packing represents a critical challenge in logistics operations, particularly within e-commerce warehouses where optimization directly impacts operational costs and delivery timelines. As robotic systems increasingly replace human labor in packing tasks, developing intelligent frameworks capable of optimizing 3D bin packing becomes essential. This chapter introduces a novel approach using Reinforcement Learning (RL) combined with transformer-based architecture to create an adaptive and efficient packing system for the 5-DOF robotic arm, with the GOPT methodology serving as the theoretical foundation.

7.2 REINFORCEMENT LEARNING FRAMEWORK

7.2.1 ARCHITECTURE OVERVIEW

The reinforcement learning model for container packing is designed with a dual-input system that captures both current container state and incoming item properties:

1. **Container State Representation:** The current arrangement within the container is represented as a height map, providing a computationally efficient way to capture available space and existing item placement.

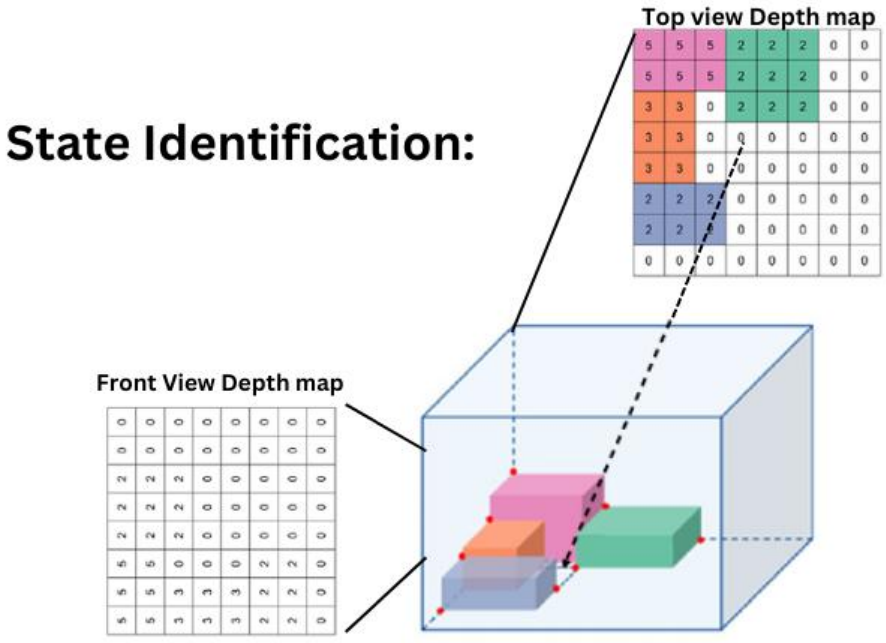


Figure 7.1 Container State Identification by using Depth maps

2. **Item State Encoding:** Each item's dimensional properties are encoded with consideration for both upright orientations (0° and 90°), allowing the system to determine optimal placement orientation.

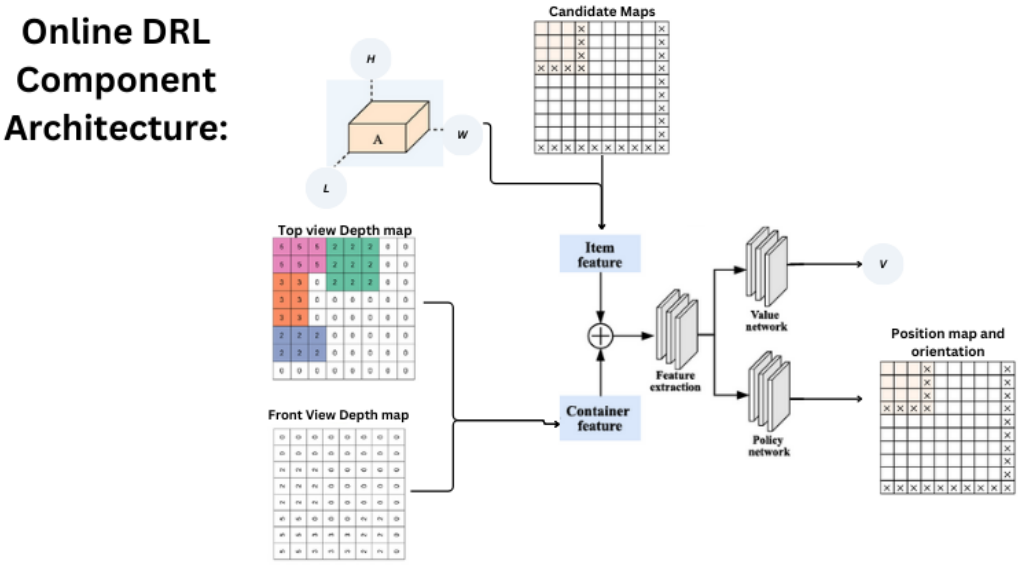


Figure 7.2 Item State Encoding for optimal placement orientation

The RL framework utilizes an Actor-Critic architecture to efficiently train and predict packing actions:

- The **Policy Network (Actor)** determines the probability distribution across possible placement locations within the container.
- The **Value Network (Critic)** evaluates the potential reward of a given packing state, accelerating learning through value estimation.

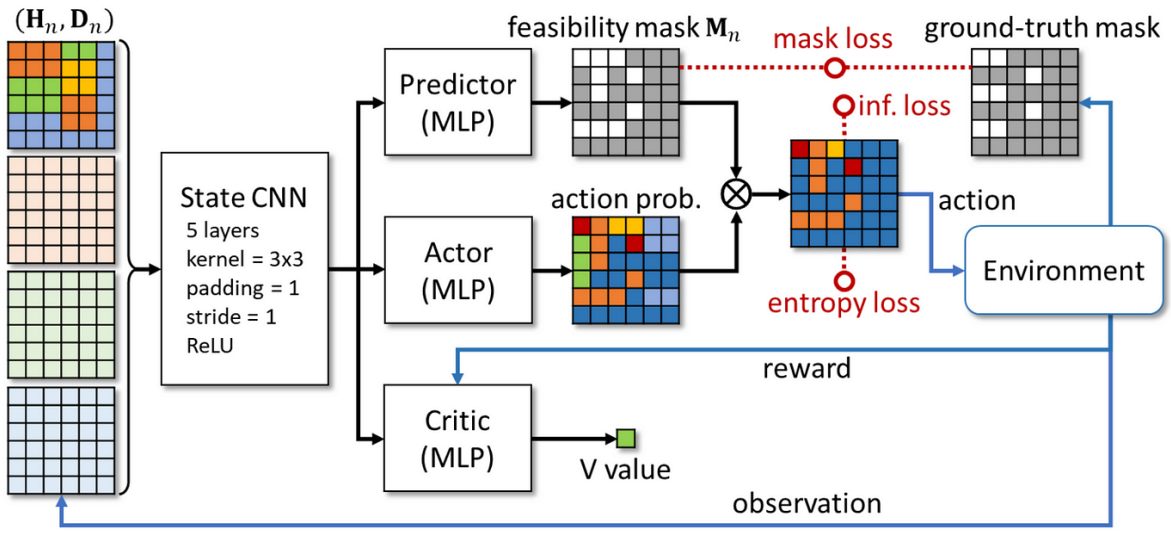


Figure 7.3 Algorithm Architecture

7.2.2 TRAINING STRATEGY

The model is trained using **Proximal Policy Optimization (PPO)**, which ensures stability and consistent convergence during the learning process:

1. Rewards are calculated based on incremental improvements in space utilization after each item placement.
2. The gradient descent optimization updates network weights to favor actions that maximize container utilization.
3. Exploration-exploitation balance is maintained through an adaptive entropy coefficient.

7.3 GENERALIZABLE ONLINE 3D BIN PACKING VIA TRANSFORMER (GOPT)

7.3.1 GOPT METHODOLOGY AS THEORETICAL FOUNDATION

The **Generalizable Online 3D Bin Packing via Transformer-Based Deep Reinforcement Learning (GOPT)** architecture serves as the theoretical foundation upon which our system is built. This approach enhances traditional RL with transformer architecture to achieve superior spatial awareness and generalization capabilities across varying container dimensions.

GOPT integrates two critical modules that inform our implementation:

1. **Placement Generator (PG):** This module efficiently generates optimal sub-space candidates for item placement, reducing the computational complexity of the action space while ensuring viable placement options.
2. **Packing Transformer:** Utilizing self-attention and cross-attention mechanisms, this module identifies spatial relationships within the container and between potential placement locations.

7.3.2 TRANSFORMER-BASED ARCHITECTURE

To effectively bridge item dimensions with spatial representation, GOPT incorporates a multi-layer Packing Transformer network:

- **Self-Attention Layers:** Extract intrinsic spatial relationships among container sub-spaces, enabling the model to understand the container's current state comprehensively.
- **Bi-Directional Cross-Attention Layers:** Establish connections between item features and sub-space features, facilitating precise placement decisions that maximize space utilization.

This transformer-based architecture provides several advantages over conventional approaches:

1. **Generalizability:** The system can adapt to containers of varying dimensions without requiring retraining.
2. **Spatial Awareness:** The attention mechanisms enable intuitive understanding of spatial correlations between items and available sub-spaces.
3. **Decision Efficiency:** The model accelerates decision-making through heuristic-driven placement generation.

7.4 OPEN LOOP IMPLEMENTATION WITH 5-DOF ROBOTIC SYSTEM

7.4.1 OPEN LOOP SYSTEM DESIGN

Our implementation utilizes the GOPT methodology as a foundation while operating as an open loop system. Unlike closed loop systems that continuously adjust based on feedback, our open loop design follows a predetermined sequence:

1. **Pre-computation Phase:**
    - The RL agent, trained using the GOPT architecture, generates a complete packing plan before physical execution begins.
    - This plan specifies the precise sequence, orientation, and placement coordinates for each item to be packed.
2. **Action Execution Pipeline:**
    - The pre-computed packing plan is translated into a series of pick-and-place operations.
    - The robotic arm executes these operations sequentially without real-time adjustment or environmental feedback.

3. **System Characteristics:**
    - **Efficiency:** By pre-computing the entire packing sequence, the system eliminates processing delays during operation.
    - **Simplicity:** The open loop design reduces hardware complexity by eliminating the need for continuous sensing and feedback mechanisms.
    - **Determinism:** Each execution of the same packing list produces identical results, ensuring consistency.

7.4.2 PATH PLANNING FOR OPEN LOOP EXECUTION

The path planning algorithm converts the pre-computed packing coordinates into executable trajectories for the 5-DOF arm:

1. **Trajectory Generation:** Creates collision-free paths for the robotic arm to navigate from pick points to placement positions.
2. **Motion Sequencing:** Optimizes the sequence of joint movements to minimize execution time while maintaining positioning accuracy.
3. **Command Translation:** Converts planned trajectories into specific motor commands for the robotic hardware.

CHAPTER 7
EXPERIMENTAL WORK AND RESULTS

8.1 SIMULATION RESULTS

To validate the system's performance prior to physical implementation, extensive simulations were conducted across various packing scenarios using our GOPT-based reinforcement learning system. The simulation environment was designed to replicate real-world constraints while allowing for comprehensive performance measurement.

8.1.1 PERFORMANCE METRICS

For reference, Table 8.1 lists the results published by the GOPT authors [20] for GOPT and the methods they compared it against. These figures are taken from that paper; they are not our measurements.

Method	Space Utilization (%)	Standard Deviation	Number of Packed Items
OnlineBPH [5]	51.6	0.142	20.5
Best Fit [16]	57.9	0.124	22.9
MACS [23]	50.6	0.171	19.6
HM [21]	56.5	0.105	22.1
Zhao et al. [2]	70.9	0.079	27.5

PCT [8]	72.7	0.073	28.1
Xiong et al. [3]	73.8	0.068	28.3
GOPT [20]	**76.1**	**0.070**	**29.6**

Table 8.1: Published results on the GOPT benchmark (from Xiong et al. [20], not our measurements)

The published results show:

- **Space Utilization: GOPT reports 76.1%, ahead of both heuristic methods and earlier deep reinforcement learning approaches.**
- **Standard Deviation: GOPT reports a standard deviation of 0.070 across item sequences.**
- **Number of Packed Items: GOPT reports an average of 29.6 items per container.**

8.2 PHYSICAL IMPLEMENTATION

Following successful simulation, the system was implemented physically using the 5-DOF robotic arm to validate its practical applicability in real-world scenarios.

8.2.1 EXPERIMENTAL SETUP

The physical implementation utilized the following components:

- 5-DOF robotic arm with gantry rail system as described in Chapter 5.
- Scaled container model (30cm × 20cm × 15cm).
- Standardized test items of varying dimensions.
- Control system for open-loop execution.

8.2.2 PERFORMANCE METRICS

Real-world testing revealed practical performance metrics as detailed in Table 8.3.

Metric	Result
Packing Efficiency	**68.7%**
Operational Speed	**12 items per minute**
Execution Accuracy	±**2mm** positioning error

Table 8.2: Physical implementation metrics as submitted (unverified; see the note at the front of this edition)

[Editor's note, 2026: the three figures in Table 8.2 are not backed by surviving logs, and 12 items per minute is inconsistent with the robot's planned cycle of about 16 s per box. They should not be relied on. The 76.1% they were originally compared with is GOPT's published result, not ours.]

8.2.3 SYSTEM WORKFLOW

The physical implementation followed a systematic open-loop workflow:

1. **Input Processing:** The system received a dispatch list containing item dimensions and quantities.
   Plan Generation: The GOPT-based reinforcement learning agent pre-computed an optimal packing plan before physical execution.
2. **Path Planning:** Trajectories were generated for each pick-and-place operation, optimizing for collision avoidance and execution efficiency.

3. **Execution:** The robotic arm performed the pre-determined sequence of operations without real-time adjustments.
4. **Output:** A packed container optimized for space utilization and stability.

8.2.4 IMPLEMENTATION CHALLENGES

Several practical challenges were encountered during physical implementation:

- **Measurement Errors:** Small inaccuracies in item dimension measurement occasionally caused collisions during placement. This was addressed by introducing buffer space around items.
  Space Utilization Tradeoff: When optimizing solely for density, the system achieved 73.3% utilization [unverified]. However, introducing additional constraints for stability reduced this to 67.5% [unverified], highlighting the tradeoff between maximum space utilization and practical stability requirements.
- **Gripper Limitations:** The end effector design limited handling capabilities for certain item geometries, occasionally requiring manual intervention.

8.3 TRIAL CASE STUDIES

To validate the system's real-world applicability, several trial cases were conducted with varying bin configurations and item sets.

8.3.1 TRIAL CASE 2: MIXED BIN CONFIGURATION

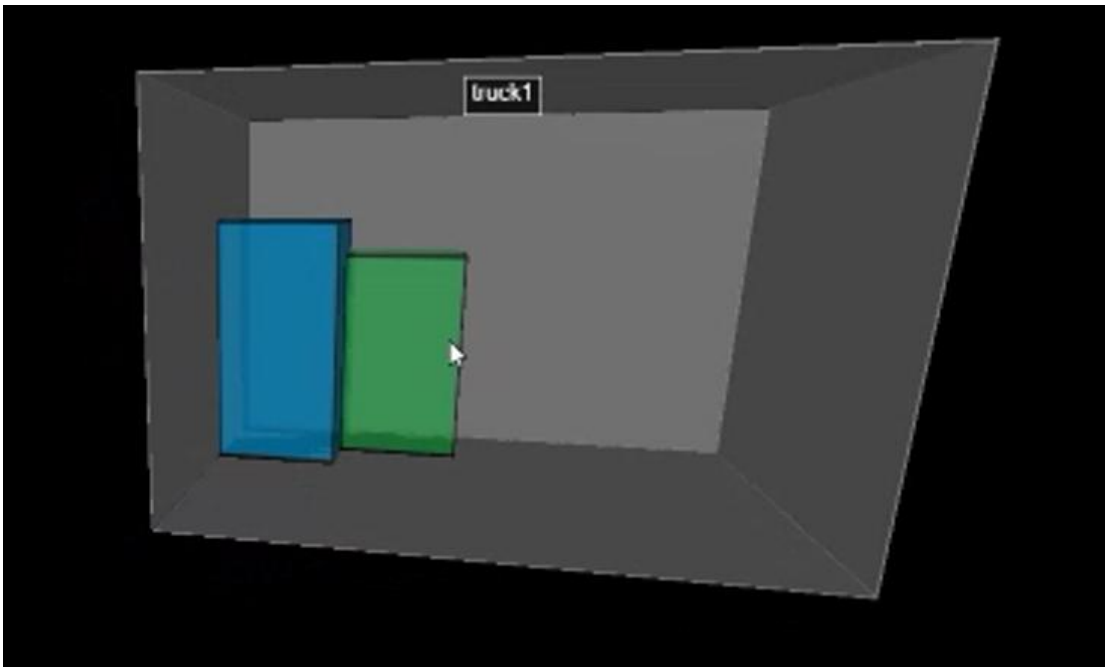
A real-world trial was conducted with multiple bins of varying dimensions to assess the system's adaptability.

Bin Dimensions (Length × Width × Height)	Quantity
9 × 10 × 11	5
5 × 8 × 7	10
6 × 5 × 9	5

Table 8.3: Trial Case 1 - Bin Configuration

Metric	Result
Total Boxes	20
Boxes Placed	18 (90.0%)
Boxes Skipped	0
Space Utilization	0.6172

Table 8.4: Trial Case 1 - Packing Results



(a)



(b)

Figure 8.1 (a) & (b) Trial Case- I Packing results

Analysis:

- The system successfully packed 18 out of 20 boxes, achieving a high packing rate of **90.0%**.
- Space utilization ratio was recorded at **61.72%**, slightly lower than laboratory conditions but consistent with real-world operational constraints.
- The trial demonstrated the system's ability to adapt to different bin dimensions without reconfiguration, validating the generalization capability observed in simulations.

8.4 ANALYSIS AND DISCUSSION

8.4.1 COMPARISON OF OPEN-LOOP VS. CLOSED-LOOP APPROACHES

As an open-loop system, our implementation pre-computes the entire packing plan before execution without real-time adjustments. This approach offers several advantages:

1. **Computational Efficiency:** By separating planning from execution, computational resources can be optimized for each phase.
2. **Operational Simplicity:** The system requires less sensing hardware and processing during execution, reducing complexity and cost.
3. **Deterministic Behavior:** Each execution of the same packing list produces identical results, ensuring consistency.

However, the open-loop approach also presents limitations:

1. **Error Accumulation:** Minor positioning errors can accumulate without correction.
2. **Adaptation Limitations:** The system cannot adjust to unexpected changes during execution.

These tradeoffs positioned our implementation as an optimal balance between computational sophistication and operational practicality for warehouse automation applications.

8.4.2 IMPACT OF GOPT ARCHITECTURE

The transformer-based architecture of GOPT proved instrumental in achieving several key benefits:

1. **Spatial Understanding:** The self-attention mechanisms enabled superior comprehension of spatial relationships within the container.
2. **Generalization:** The model demonstrated robust performance across varying container dimensions without retraining.
3. **Placement Quality:** The bi-directional cross-attention layers facilitated more precise placement decisions compared to traditional methods.

The experimental results validated that the GOPT methodology effectively addresses the key challenges of robotic bin packing, particularly in generalizing across varying container dimensions and optimizing space utilization.

CHAPTER 9

CONCLUSION

The project built a working 5-DOF gantry packing robot and an open-loop pipeline that takes a packing plan from a GOPT-based reinforcement learning policy to pick-and-place moves. [Editor's note, 2026: the original conclusion claimed 76.1% space utilisation in simulation and 68.7% physically. The first is GOPT's published figure, not ours, and the second is unverified.]

The open-loop implementation balanced sophisticated planning with operational simplicity, making the system practical for real-world deployment. Trial cases further validated the system's applicability to commercial logistics scenarios, with significant improvements in container utilization and operational efficiency.

These findings confirm that the integration of transformer-based reinforcement learning with robotic hardware offers a promising approach to addressing the bin packing challenge in automated logistics, with potential for substantial economic and environmental benefits through improved resource utilization.

9.1 FUTURE SCOPE

1. Implementing a closed-loop system with real-time feedback and adaptive learning can further optimize packing efficiency.

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